

COSC 3337

Include information here about week 0 (Think homework 0, and what you will need to install for the class). Information about: CMAPs, Homeworks, Grading Structure (use one from this semester), In-class labs, Exam Structure (format, etc...)

Sections:

1. Week 0 - Thy
2. CMAPs - Thy
3. CIAs- Buu
4. Homework-Buu
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6. Practices
7. Grading Structure - Thy
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9. Exam structure-Buu

1. Week 0 - Thy
 - What software is needed for this course?

We use Python throughout this course, and we recommend you install Jupyter and use Anaconda environments. As for CMAP assignments you can install Cmap software by IHMC, which you can download from [Cmap](#)

- Are there any alternatives?

You can use Google Colab for coding assignments such as your homeworks, labs, practices, and Cmap Cloud for CMAPs [Cmap Cloud](#) or draw.io

- What are the differences between Google Colab and Jupyter Notebook?

The biggest difference between Google Colab and Jupyter Notebook is that Google Colab is cloud-based, and Jupyter Notebook works offline. Google Colab offers pre-installed packages, which makes it quicker to start your project. However, it is slower than Jupyter Notebook when loading large datasets, so we recommend you use Jupyter Notebook for your homework.

- What are the first

2. CMAPs - Thy
 - What is CMAP?

CMAP is short for Concept Map. For these assignments, you need to gather information about the assigned topic and visualize it by connecting related concepts with linking phrases.

You are free to do as much research on the topics as you want. We recommend you go through the PowerPoint slides.

You will need to submit a .cxl and a .pdf file for each CMAP.

- How are CMAPs graded?

CMAPs are graded based on the number of concepts and the highest hierarchy, following this formula: Number of concepts + 5 × Highest hierarchy.

- How should I do a CMAP?

First, gather detailed information about the topic. Try to answer questions such as What, Why, Where, When, and How. Then create a concept map based on that information. Make sure to submit a .cxl and a .pdf file. You can also estimate your grade using the Concept Map Grader using this link: [Concept Map Grader](#)

- How many CMAPs are there?

We have 4 CMAPs that make up 4% of your total grade. CMAP1 is due on Saturday June 13th at 11:59 PM, CMAP2 is due on Saturday June 20th at 11:59 PM, CMAP3 is due on Saturday June 27th at 11:59 PM, CMAP4 is due on Sunday July 5th at 11:59 PM

- What are the topics for CMAPs?

The topic for each CMAP is based on the topics covered in your next exam, which happens 1-2 days after the CMAP is due. The topic for CMAP 0 is Introduction to Data Science. The grade of this assignment will not count towards your final grade, so you should use this as a practice for your upcoming CMAP assignments.

3. CIAs- Buu

What is the CIA?

- A CIA is a team assignment in which three students use ChatGPT to develop a 20 question quiz, plus a 21st question that rates the quiz itself. Two CIAs are submitted across the semester, before each exam. CIAs are explicitly designed to serve as exam prep, so do not treat them as throwaways.

What you Build

- Each team produces a 20 question quiz with a strict question mix:
 - 5 multiple choice questions
 - 5 true/false questions
 - 5 word-answer questions
 - 5 fill in the blank questions
 - Question 21: a 1-10 rating of the quiz itself

At least one question should involve coding in Python. You also submit a short video explaining the correct answers. The text file is converted to a Canvas QTI .zip using the NYIT converter([Canvas Exam Converter | Information Technology Services | New York Tech](#)), and the .txt, .zip, and video are uploaded to Canvas

Naming Convention

- Question file: Cougarid1_Cougarid2_CIA1.txt
- Video: Cougarid1_Cougarid2_CIA1_V
- One submission per team - either student can upload all the file.

Template Format(quick reference)

The student template uses a simple text syntax that the NYIT converter understands, Example(asterisk marks the correct answer for MC and T/F; bracketed asterisks mark corrections for multi-select; the * lines under fill-in list all acceptable answer):

Multiple Choice: Each choice needs to start with a lower case alphabet, a, b, c, d, etc. with a close parenthesis. The correct choice is designated with an asterisk.

1. What is 2+3?

- a) 6
- b) 1

- *c) 5
- d) 10

Multiple-answers: Each choice use ☐ or ☒ to make the incorrect answers and [*] for the correct answers.

1. Which of the following are dinosaurs?
- ☐ Woolly mammoth
 - [*] Tyrannosaurus rex
 - [*] Triceratops
 - ☐ Smilodon fatalis

Short-answer (fill-in-the-blank): The correct answers will use an asterisk followed by one or more spaces or tabs followed by an answer.

1. Who lives at the North Pole?
- * Santa
 - * Santa Claus
 - * Father Christmas
 - * Saint Nicholas
 - * Saint Nick

21. Rate this Quiz done by Cougarid1_Cougarid2. Select one option only.

- [*] 1
- [*] 2
- [*] 3
- [*] 4
- [*] 5
- [*] 6
- [*] 7
- [*] 8
- [*] 9
- [*] 10

4. **Homeworks-Buu**

Four homeworks are due on the same days as the four exams (Mondays at 11:59 PM, except HW 4, which is due Monday, July 6). Plan to have each homework done before the exam, not after. Monday is going to be a long day either way.

Homework Schedule

Homework	Due Date	Topic
HW 1	Monday June 15, 11:59 PM	Data cleaning, exploration, regression

HW 2	Monday June 22, 11:59 PM	Dimensionality reduction, classification
HW 3	Monday June 29, 11:59 PM	Model evaluation, clustering
HW 4	Monday July 6, 11:59 PM	Neural networks, association rules

Submission Tips

- All code must be a Jupyter notebook (.ipynb). No .py files for homework.
- Run all cells before saving so outputs are visible to the grader.
- Add markdown cells between code blocks to explain your reasoning
"Interpretation" is a graded criterion.
- Label every plot (title, x-axis, y-axis, legend). Unlabeled plots cost you the visualization criterion.
- No late submissions accepted. None. Submit even if incomplete.

5. Labs

6. Practices - Vincent

What these are:

Exercises allow students to train their ability to utilize python for specific scenarios using different methods depending on the scenario. Some practices will help students learn the basic necessities for python coding that is used at every part of the coding portion within the course.

The practices are divided into multiple different parts, there being 13 individual practices, each part pertaining to their own methodology. Each practice helps students learn how to use a specific data science topic, from importing, to initializing, to actually using libraries.

The practices range from a beginner's level of python understanding, using core libraries and functions for data science to each main topic within the course - showing students what libraries need to be imported, how to utilize the libraries, what variables and parameters goes with what functions, and overall syntax for each topic.

They make up for an additional 1% of the student's grade if done.

How to do them:

Open a Jupyter notebook or Google Colab and follow essentially copy what is in the practice `\*.html` and/or `\*.pdf`. When copying the code cells, execute them and watch what happens in response.

For better understanding of what specific parts of the code does, experiment with the parameters and variables within the function calls. Play around with what the code does. It is

best to try different things with what is given in the files, to change some values around and see how they affect the outcomes.

It is highly recommended to finish these practices as soon as possible, but not just copying and submitting them as is, instead actually see what specific segments of code does as to understand how things are actually working.

7. Grading Structure - Thy

The final numeric grade is computed based on the student's performance in weekly assignments and exams/quizzes. The final numeric grade for the course will be determined based on 4 exams, 4 homework assignments, 3 Labs and In-class, 4 CMAPs, 2 CIAs, Attendance, and additional points if applicable:

- 4 exams: 68% of the total grade, each is worth 17%. Exam 1 is on Monday, June 15th, Exam 2 is on Monday, June 22nd, Exam 3 is on Monday, June 29th, Exam 4 is on Tuesday, July 7. All exams will be taken in class at the same time as the lecture, from 8:00 AM to 10:00 AM.
- 4 homework assignments: 12%. Hw1 is due on Monday June 15th, at 11:59 PM, Hw2 is due on Monday June 22nd, at 11:59 PM, Hw3 is due on Monday June 29th, at 11:59 PM, Hw4 is due on Monday July 6th, at 11:59 PM.
- 3 Labs and Inclass: 10%. Lab1 is due on Sunday June 21st at 11:59 PM, lab2 is due on Sunday June 28th at 11:59 PM, lab3 is due Sunday July 5th at 11:59 PM
- 4 CMAPs: 4%. CMAP1 is due on Saturday June 13th at 11:59 PM, CMAP2 is due on Saturday June 20th at 11:59 PM, CMAP3 is due on Saturday June 27th at 11:59 PM, CMAP4 is due on Sunday July 5th at 11:59 PM
- 2 CIAs (ChatGPT Interactive Activities): 3%. CIA1 is due on Tuesday June 16th at 11:59 PM, CIA2 is due on Tuesday June 23rd at 11:59 PM.
- Attendance: 3%.
- Additional points, if applicable:

1. Practices (13 in total): 1%

2. 3 points credit towards the final exam, per completed individual Project.

3. 7 points credit towards the final exam , from Participation in Data Science Show case.

8. In-class labs

9. Exam Structure-Buu

There are four exams, each worth 17%, held in class at the regular meeting time (8:00–10:00 AM). Three of the four exams fall on Mondays; the final exam (Exam 4) is on Tuesday, July 7. Exams are closed-book and combine conceptual questions, mathematical reasoning, and short Python coding.

Exam Schedule

Exam	Date	weight	Coverage
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Exam 1	Monday June 15	17%	Week 1: intro to DS, data cleaning, exploration, similarities & distances, linear & logistic regression
Exam 2	Monday June 22	17%	Week 2: dimensionality reduction, KNN, decision trees, random forests, Naive Bayes
Exam 3	Monday June 29	17%	Week 3: model evaluation, ridge/lasso, SVM, K-means, hierarchical clustering, heatmaps
Exam 4	Tuesday July 7	17%	Weeks 4–5 + cumulative: storytelling, DBSCAN, cluster validity, neural networks, Apriori

Exam 4 is the final and can be padded with bonus credit: 3 points per completed individual project and 7 points for participation in the HPDSI Showcase. If you want either bonus, plan for it at the start of the term, not the last week.

How to Prepare

- Redo every homework and lab, exam questions follow the same patterns.
- Use the CMAP you built that week as a study guide for that week's exam.
- Each CIA you build is by design a 20 question practice quiz on the exam topic. Use them.
- Bring two pencils, an eraser, and a non-programmable calculator. No phones, no headphones.
- Arrive 5 minutes early. Latecomers do not receive extra time.
- Makeup & Grade Objection Rules
- No makeup policy: Missed exams are scored zero unless you have an excused absence with full documentation submitted at least two working days BEFORE the exam. Documentation submitted on the day of, after, or within one day of the exam results in a zero. Even with valid documentation, makeup is not guaranteed.
- 1 Day grade objection rule: You have exactly one day from the end of the class in which a graded exam or assignment was returned/posted to object to the score. Objections must be emailed to both the TA and the instructor. After 24 hours, the grade is final.